Low-Cost Optical Sensor Reliability Under Simulated Real-World Disturbances

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Portfolio Technical Report

Project context: This report consolidates the methods and final results of a student-led bench-top experiment completed through Johns Hopkins Biomedical Engineering Innovation. It is a portfolio summary of the completed project, not an independent or peer-reviewed publication.

Project Summary

This project evaluated how common real-world disturbances affect the raw infrared (IR) output of a low-cost MAX30102 optical sensor. An Arduino Uno R4 Minima was used to collect raw sensor readings under five bench-top conditions: baseline, motion, loose contact, ambient light, and environmental disturbance. The goal was to evaluate signal reliability rather than calculate BPM or assess clinical accuracy.

Research Question

How do motion, loose sensor contact, ambient light, and environmental disturbance affect the reliability of raw infrared output from a low-cost optical heart-rate sensor?

Hypothesis

Disturbance conditions were expected to reduce raw IR signal reliability relative to the stable baseline, appearing as greater variability and/or more invalid or unstable readings.

Experimental Setup

Component	Role
Arduino Uno R4 Minima	Sensor interface and serial data collection
GY-MAX30102 / MAX30102	Low-cost optical sensor
Arduino IDE / Serial Monitor	Condition labeling and raw IR output recording
Bench-top test surface	Consistent non-human testing surface
Phone flashlight / spacer / manual movement	Simulated disturbance conditions

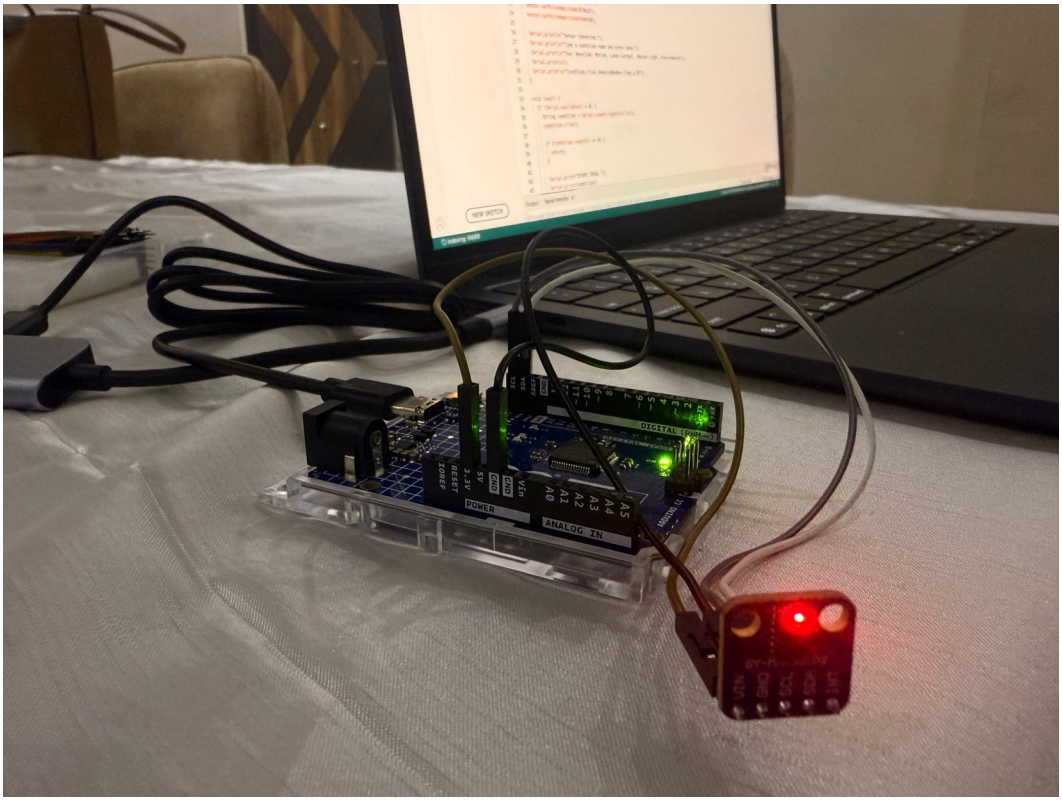


Figure 1. Arduino Uno R4 Minima and MAX30102 optical sensor setup used during the project.

Methods

The experiment used a bench-top design rather than human-subject testing. The independent variable was the testing condition. Five conditions were evaluated: baseline, motion, loose contact, ambient light, and environmental disturbance. Each condition was repeated three times.

The original Arduino sketch accepted the condition name through the Serial Monitor and recorded the condition, trial number, reading number, elapsed time, and raw IR value. The preserved code collected 30 raw IR readings per trial with a programmed 500 ms interval between readings. The same Arduino board, sensor, acquisition code, and general recording procedure were maintained across conditions.

Reliability Measures

- Mean raw IR reading - overall signal level.
- Standard deviation - variation in the recorded signal.
- Range - maximum minus minimum recorded value.
- Invalid/unstable count - final count preserved from the completed project analysis.

Raw IR values are reported in arbitrary units (a.u.). They are not BPM, SpO2, diagnostic measurements, or a direct measure of clinical accuracy.

Results

Condition	Mean (a.u.)	SD (a.u.)	Range (a.u.)	Unstable
Baseline	593.66	22.66	151	2
Motion	306.83	13.08	69	3
Loose Contact	78,571.48	48,943.88	130,209	6
Ambient Light	332.03	35.64	150	10
Environmental	400.81	1,320.50	11,920	21

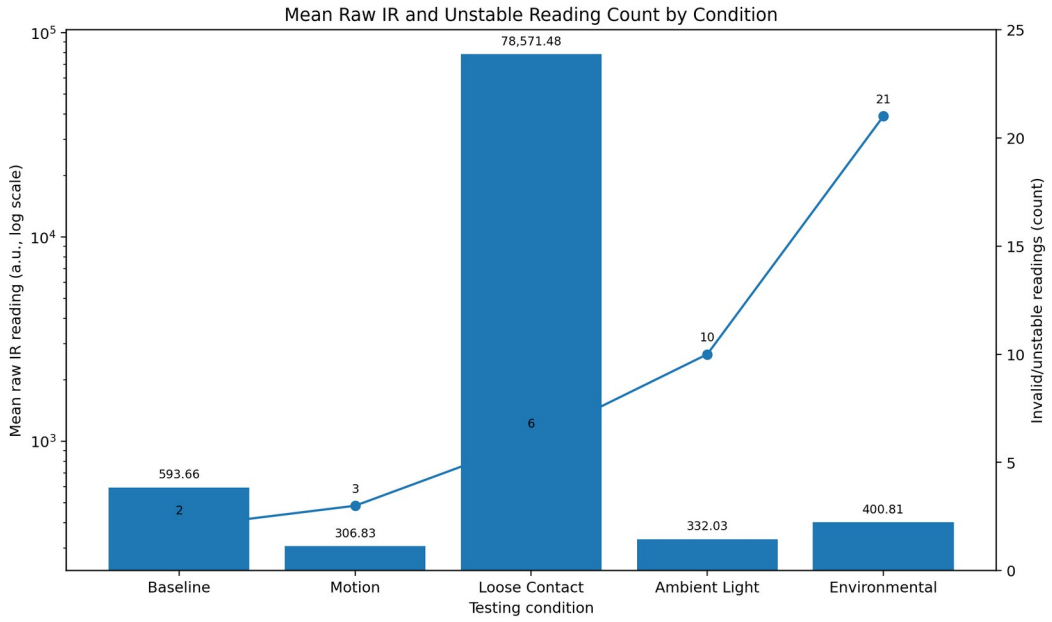


Figure 2. Mean raw IR output and final invalid/unstable reading count by testing condition.

Interpretation

Loose contact produced the strongest overall signal variation. Its mean raw IR value (78,571.48 a.u.), standard deviation (48,943.88 a.u.), and range (130,209 a.u.) were substantially larger than the other conditions. This result is consistent with the importance of stable optical geometry: changing the sensor-to-surface relationship can substantially alter the amount of light reaching the detector.

Environmental disturbance produced the highest invalid/unstable reading count (21), followed by ambient light (10). Baseline had the fewest unstable readings (2), while motion had the lowest standard deviation and range. Because different disturbance conditions affected different reliability metrics, the original hypothesis was only partially supported.

Limitations

- Only three trials were completed per condition.
- Disturbances were applied manually rather than with instrument-controlled magnitude.
- Only one optical sensor module was evaluated.
- The experiment measured bench-top raw IR reliability, not human heart-rate accuracy.
- The archived project materials preserve summary results but not the original raw Serial Monitor logs.

Future Work

A stronger follow-up study could increase the number of trials, standardize motion with a mechanical platform, control sensor-to-surface spacing with a fixed holder, measure light intensity and distance, compare multiple sensor modules, and test signal-quality or artifact-detection algorithms. Any later follow-up should be documented as a separate phase rather than retroactively changing the context of the original BMEI project.

Conclusion

The project showed that raw optical-sensor reliability depended strongly on test conditions. Loose contact produced the greatest overall signal variability, while environmental disturbance produced the greatest number of invalid or unstable readings. The project demonstrates an engineering principle relevant to wearable sensing: reliable downstream measurements depend on stable upstream signal acquisition.

References

- Bent, B., Goldstein, B. A., Kibbe, W. A., & Dunn, J. P. (2020). Investigating sources of inaccuracy in wearable optical heart rate sensors. npj Digital Medicine, 3, 18.
- Charlton, P. H., et al. (2022). Wearable photoplethysmography for cardiovascular monitoring. Proceedings of the IEEE, 110(3), 355-381.
- Castaneda, D., Esparza, A., Ghamari, M., Soltanpur, C., & Nazeran, H. (2018). A review on wearable photoplethysmography sensors and their potential future applications in health care. International Journal of Biosensors & Bioelectronics, 4(4), 195-202.

Repository Note

The accompanying repository includes the original Arduino acquisition sketch, final condition-level summary data, an actual hardware photograph, and a small post-course plotting script that reproduces the portfolio results figure. No missing raw data have been reconstructed.